



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO RAFI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

before reaching Gandhinagar.

- Roads utilized include Salem - Kochi Highway, Omalur-Sankagiri Road, and Mettur Main Road (SH 86), all of which are major routes linking small towns and industrial areas.

2. Typical Weather Conditions and Weather-Related Hazards:

- Tamil Nadu generally experiences three climatic conditions: summer (April-June), monsoon (July-September and October-November), and winter (December-March).
- Potential weather-related hazards include heavy rainfall and subsequent flooding during the monsoon seasons, which can affect visibility, road conditions, and potentially cause landslides in hilly areas near Danishpet.

3. Traffic Patterns and Congestion-Prone Areas:

- Expect peak traffic hours in the mornings (7:30 AM - 9:30 AM) and evenings (5:00 PM - 8:00 PM).
- Congestion is commonly found in commercial hubs around Omalur and Salem due to local market activities and vehicular density.
- Roads near schools or industrial areas in Sangagiri can also experience short-term congestion.

4. Assessment of Road Quality and Infrastructure:

- The road surface is generally well-maintained on highways and major roads, with regular signage and lighting.
- Potholes and wear may be present on smaller roads, especially in rural stretches or areas frequently hit by rainfall.
- Be cautious of narrow bridges and sharp turns, specifically on less frequented rural stretches between Desavilakku and Danishpet.

5. Suggestions for Alternative Routes for Emergencies:

- In case of severe congestion or roadblock on the main route, consider using the NH 544 that bypasses Salem, or alternate state highways connecting through adjacent towns.
- Availability of side roads in rural stretches can be sparse; hence, plan and identify key exit routes before and through the journey for emergencies.

6. Local Regulations Affecting Hazardous Material Transport:

- Transporting hazardous materials requires adherence to Tamil Nadu's guidelines and securing necessary permits.
- Time restrictions are sometimes imposed in residential and commercial zones, prohibiting heavy vehicle movement during peak hours.

7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

- While specific databases should be used for detailed research, reports suggest occasional accidents involving vehicular collisions or spillages, typically attributed to human error or weather conditions.
- Regularly updated regulations and training have reduced frequent occurrences over recent years.

8. Environmental Considerations and Sensitive Areas:

- The route traverses through regions of agricultural importance and some forest reserves, particularly near Danishpet.

- Reducing emissions and observing strict waste disposal protocols are vital to protecting these environments.

9. **Communication Coverage and Potential Dead Zones:**

- Most of the route provides adequate mobile network coverage; however, expect potential dead zones in sparsely populated rural areas, especially near forest fringes around Danishpet.
- Drivers should be equipped with offline navigation and emergency contact tools.

10. **Estimated Emergency Response Times:**

- Urban areas like Omalur and Salem offer faster emergency response (approximately 15-30 minutes) due to proximity to services.
- Rural sections may experience delayed response times, possibly extending to 45 minutes to an hour, given terrain and remoteness.

11. **Overall Summary of Risk Assessment:**

- The route presents manageable risks for transporting heavy and hazardous materials, provided adherence to weather advisories, road condition reports, and local transport regulations.
- Critical risks include monsoon-related issues, human factors during nighttime driving, and environmental protection compliance.
- Regular updates and en route communication are necessary to mitigate unforeseen disruptions.

Drivers and transport coordinators must follow these insights for safer and efficient transit along this crucial route in Tamil Nadu.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43961, 77.87341	15 KM/Hr
1	Turn	High	11.44024, 77.87528	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45847, 77.85140	30 KM/Hr
5	Turn	Medium	11.45898, 77.85135	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	Medium	11.45543, 77.81740	30 KM/Hr
8	Turn	High	11.45631, 77.81668	15 KM/Hr
9	Turn	High	11.49383, 77.88442	15 KM/Hr
10	Turn	Medium	11.56083, 77.89921	30 KM/Hr
11	Turn	Medium	11.56180, 77.90019	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	Medium	11.58254, 77.90656	30 KM/Hr
13	Turn	Medium	11.63867, 77.94841	30 KM/Hr
14	Turn	Medium	11.64217, 77.95200	30 KM/Hr
15	Turn	Medium	11.64712, 77.95118	30 KM/Hr
16	Turn	High	11.67858, 77.96834	15 KM/Hr
17	Turn	High	11.70233, 77.97869	15 KM/Hr
18	Turn	Medium	11.71394, 77.99572	30 KM/Hr
19	Turn	Medium	11.71916, 78.01117	30 KM/Hr
20	Turn	Medium	11.72559, 78.01758	30 KM/Hr
21	Turn	Medium	11.73787, 78.03528	30 KM/Hr
22	Turn	High	11.73651, 78.03868	15 KM/Hr
23	Turn	Medium	11.74257, 78.04708	30 KM/Hr
24	Blind Spot	Blind Spot	11.87104, 78.08356	10 KM/Hr
25	Turn	Medium	11.87071, 78.08450	30 KM/Hr
26	Turn	Medium	11.87089, 78.08558	30 KM/Hr
27	Turn	Medium	11.87249, 78.08707	30 KM/Hr
28	Turn	Medium	11.86893, 78.12084	30 KM/Hr
29	Turn	Medium	11.86491, 78.12535	30 KM/Hr
30	Turn	High	11.85820, 78.13580	15 KM/Hr
31	Turn	Medium	11.86652, 78.13879	30 KM/Hr
32	Turn	High	11.88411, 78.14478	15 KM/Hr
33	Turn	Medium	11.90581, 78.15292	30 KM/Hr
34	Turn	High	11.90760, 78.15205	15 KM/Hr
35	Turn	Medium	11.91226, 78.15594	30 KM/Hr
36	Turn	Medium	11.91312, 78.15616	30 KM/Hr
37	Turn	Medium	11.91519, 78.15837	30 KM/Hr
38	Turn	High	11.92454, 78.17073	15 KM/Hr
39	Turn	Medium	11.92820, 78.18040	30 KM/Hr
40	Turn	Medium	11.92797, 78.18165	30 KM/Hr
41	Turn	Medium	11.92890, 78.18331	30 KM/Hr
42	Turn	Medium	11.93569, 78.18711	30 KM/Hr
43	Turn	Medium	11.95116, 78.20647	30 KM/Hr
44	Turn	Medium	11.95200, 78.20703	30 KM/Hr
45	Turn	High	11.95258, 78.20848	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
46	Turn	Medium	11.95173, 78.21017	30 KM/Hr
47	Turn	High	11.95221, 78.21173	15 KM/Hr
48	Turn	Medium	11.95165, 78.21276	30 KM/Hr
49	Turn	Medium	11.95228, 78.21413	30 KM/Hr
50	Turn	Medium	11.95220, 78.21522	30 KM/Hr
51	Turn	Medium	11.95288, 78.21603	30 KM/Hr
52	Turn	Medium	11.95264, 78.22467	30 KM/Hr
53	Turn	High	11.95178, 78.22600	15 KM/Hr
54	Turn	Medium	11.95843, 78.23409	30 KM/Hr
55	Turn	Medium	11.96188, 78.23441	30 KM/Hr
56	Turn	High	11.96666, 78.23282	15 KM/Hr
57	Blind Spot	Blind Spot	11.96699, 78.23204	10 KM/Hr
58	Turn	High	11.97325, 78.23714	15 KM/Hr
59	Turn	High	11.97765, 78.23556	15 KM/Hr
60	Turn	High	11.98243, 78.24308	15 KM/Hr
61	Turn	High	11.98206, 78.24332	15 KM/Hr
62	Turn	Medium	11.99307, 78.25808	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
17	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
18	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
19	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
20	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
21	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
24	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
26	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
27	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
28	clinic	Kamala Clinic	11.8723673, 78.0874195	30 km/h	Medium
29	hospital	Kamala Hospital	11.8722588, 78.0899189	30 km/h	Medium
30	hospital	GJB Hospital	11.8713998, 78.0980288	30 km/h	Medium
31	clinic	Pranav Clinic	11.8712846, 78.0998167	30 km/h	Medium
32	hospital	Bethel Hospital	11.8699472, 78.1388598	30 km/h	Medium
33	hospital	Amirtham Hospital	11.981994, 78.248038	30 km/h	Medium
34	hospital	Grs Hospital	11.9824068, 78.2481763	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
25	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44024, 77.87528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49383, 77.88442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56083, 77.89921



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56180, 77.90019



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58254, 77.90656



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63867, 77.94841



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64217, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64712, 77.95118



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70233, 77.97869



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71394, 77.99572



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71916, 78.01117



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72559, 78.01758



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73787, 78.03528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03868



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74257, 78.04708



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.87104, 78.08356



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87071, 78.08450



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87089, 78.08558



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87249, 78.08707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86893, 78.12084



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95116, 78.20647



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.97325, 78.23714



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.97765, 78.23556



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.98243, 78.24308



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.98206, 78.24332



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.99307, 78.25808
